

Executive Summary

Automated Detection, Correction, and Surface Fairing for Skin-to-Muscle Registration Artifacts. Ilian Khankhalaev, Simon Fraser University. Supervisor / Project Lead: Richard Peng. September 2026.

Problem. After physical registration of a template skin onto personalised muscles, fat and bone, the 15,905-vertex skin still showed local bumps and ridges and, in 93 closed-skin faces, internal anatomy crossing the surface. Eye and mouth openings legitimately expose anatomy and had to be preserved.

Approach. Detection was developed in five milestones (umbrella Laplacian, reference-Laplacian, multi-scale/hybrid, anatomy-SDF reference, M3UM4 fusion) and frozen at M5. A triangle-triangle intersection detector became the safety metric. Repair and constrained-smoothing solvers were then studied until a pure Laplacian diagnostic showed that the constraints, not the operator, limited progress; a blocked Laplacian localised the constraints, and a small documented manual pass finished the mesh.

Key results (all recomputed offline from the saved Maya scene).

Table 1. Headline before/after values (repository intersection detector; roughness = umbrella-Laplacian norm in world units).

Metric	Pre-cleanup skin	After unified solver (_solver200)	Saved final scene
Forbidden intersecting faces (pairs)	93 (497)	0 (0)	3 (15)
All-vertex mean roughness	0.1345	0.1133	0.0965
All-vertex p99 roughness	0.681	0.666	0.402
Rough-core mean roughness (n=3,564)	0.252	0.276	0.191

Conclusion. The research question is answered: bumps are removable by simple Laplacian smoothing, and the anatomical constraint belongs in a local freeze rather than in a global weakening of the smoother. The saved final mesh is a hybrid result (automated detection and smoothing plus a documented manual pass). It still contains 3 forbidden faces by the offline detector, and no fully automatic production system is claimed; both points are treated in Sections 14 and 15 of the full report.

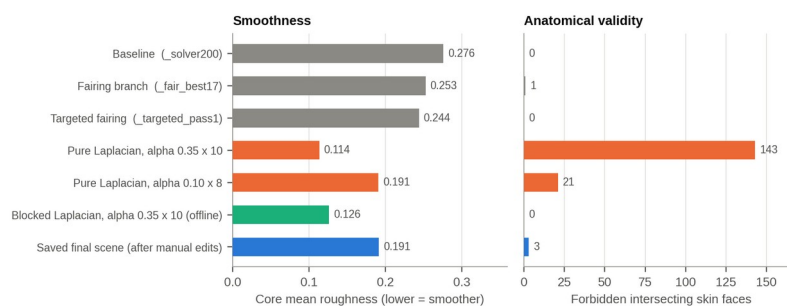


Figure 1. Smoothness against safety for the smoothers studied: rough-core mean roughness (left, lower is smoother) and forbidden intersecting faces (right). The unified solver and fairing passes are safe but leave the surface rough; the pure Laplacian is smooth but unsafe; the blocked Laplacian (offline reproduction) is both smooth and free of forbidden faces.

What is not claimed

The saved final mesh still has 3 forbidden faces by the offline detector (Maya-side confirmation pending). The blocked Laplacian is shown by offline reproduction and is not the origin of the saved final. Only one character and one registration were studied, and there is no labelled ground truth. The missing evidence is listed in Appendix E of the full report.